APQP LESSONS LEARNED TEMPLATE & GUIDE

Capture, Store, and Reuse Program Knowledge | IATF 16949 Clause 10.3 | Quality Head Agents

PURPOSE

The Lessons Learned system captures knowledge from every APQP program — problems, successes, near-misses, and innovative solutions — and feeds them into future programs. IATF 16949 Clause 10.3 requires that lessons learned from previous programs are incorporated into new APQP plans.

LESSONS LEARNED ENTRY TEMPLATE

FIELD	CONTENT / INSTRUCTIONS
LL Reference Number	Format: LL-YYYY-NNN (e.g., LL-2024-001)
Program Name / Part Number	Part name, drawing number, customer
Date of Event / Discovery	When the issue occurred or was discovered
APQP Phase	Phase 1, 2, 3, 4, or 5 where issue occurred
Category	DFMEA | PFMEA | Control Plan | MSA | Capability | PPAP | Customer | Supplier | Launch | Process
Problem Statement	Clear, factual description of what went wrong (or what worked well)
Root Cause	Why did this happen? (5-Why or Fishbone result)
Escape Point	Where was the issue detected? (Design, Process, Final Inspection, Customer)
Customer Impact	Was customer affected? PPM impact? Warranty claims?
Cost of Poor Quality (COPQ)	Scrap cost, rework cost, warranty cost, premium freight
Correction Taken	What was done to fix this specific issue?
Corrective Action (Systemic)	What systemic change prevents recurrence?
Preventive Action (Future Programs)	How should future APQP be different because of this?
Documents Updated	Which documents were updated? (PFMEA, CP, WI, Training, etc.)
Applicable Future Programs	Program types where this LL should be applied
Verified Effective?	Date verified effective: _______
Prepared by	Name and signature
Reviewed by	Quality Manager name and signature

EXAMPLE LESSONS LEARNED ENTRIES

LL-2024-001 — PFMEA / Control Plan Gap

Problem: Problem: SC for weld joint strength was identified in DFMEA but was not cascaded to PFMEA and Control Plan. Discovered during internal audit 6 weeks before PPAP submission.

Root Cause: Root Cause: No formal SC cascade verification step in APQP process. DFMEA and PFMEA were created by different teams without a cross-check.

Lesson: Lesson: Add 'SC Cascade Verification' as a mandatory step in Gate Review 3. Quality Manager must sign the SC Cascade Checklist confirming that every SC in the drawing and DFMEA appears in the PFMEA and Control Plan.

LL-2024-002 — GRR Failure at PPAP

Problem: Problem: GRR study for critical dimension measurement system showed 24% — unacceptable. PPAP submission delayed by 3 weeks while a new gauge was procured.

Root Cause: Root Cause: Gauge selected based on cost, not capability. GRR study was planned but conducted too late (only 1 week before PPAP submission date).

Lesson: Lesson: Add GRR completion for all SC gauges as a mandatory Gate 3 deliverable (not Gate 4). Minimum 4 weeks before PPAP submission. Gauge selection criteria must include GRR feasibility assessment.

LL-2024-003 — Production Trial Run Volume

Problem: Problem: PTR conducted with only 150 pcs due to raw material shortage. Cpk appeared acceptable. Post-launch, actual Cpk dropped to 1.1 — customer warranty claims.

Root Cause: Root Cause: PTR sample size was below AIAG minimum of 300 pcs. Special approval was taken without proper risk assessment.

Lesson: Lesson: No PTR with <300 pcs without written customer approval and risk assessment signed by Plant Head and Quality Director. PTR results with <300 pcs cannot be used for PPAP capability submission.

LESSONS LEARNED INTEGRATION INTO APQP

- Phase 1 Kickoff: Review LL database for similar programs — identify applicable lessons
- PFMEA Development: Incorporate LL-identified failure modes as starting point
- Gate Reviews: LL review is a standard agenda item at every gate
- Phase 4 PTR: Check if any LL relates to production trial run requirements
- Phase 5 Closure: Mandatory LL capture — minimum 3 entries per program
- Annual Review: Quality Manager reviews LL database, identifies patterns, updates SOPs